Anchored multi-omics integration and knowledge-anchored residual discovery

A never-below-the-best-view integrator, and a tool to find what the known biology misses.

H. Ryan Kim · omniomics-prototype · 2026

In brief

Combining many kinds of molecular data is widely assumed to improve cancer prediction, yet in practice a single strong data type is hard to beat. This work introduces an *anchored* framework that starts from established biological knowledge — a known high-performing data type, or a fixed textbook gene signature — so the model can never do worse than the existing standard. By analysing only the *residual* signal left after accounting for that anchor, it surfaces orthogonal biological axes that conventional data-fusion overlooks: it recovers a keratinization (basal) program in breast cancer and the field's established immunotherapy biomarkers in lung cancer. Inverting the idea, a hypothesis can itself be tested against the textbook prior, letting the framework separate genuinely new findings from those already explained by existing literature. It also makes *transportability* explicit — showing how the measurement technology (for example, RNA-seq versus microarray) can make the same biological signal look novel in one cohort and redundant in another. In short, it is a discovery engine that is honest about what is new and what the textbook already knows.

Abstract

Multi-omics integration is usually sold as "more modalities → better predictions," yet on real cancer data a strong single modality is hard to beat and naive fusion often does worse. We define an **anchored** integrator that anchors on the empirically strongest (or a fixed prior) modality and adds the others only as a non-negative gated residual, so it is provably never below its best single view and improves on it only where another modality carries orthogonal signal. Across nine TCGA-BRCA endpoints the gate adds nothing where one modality dominates and engages only on a constructed positive control — quantifying, rather than assuming, where multi-omics helps. Generalising the anchor from "best modality" to **established knowledge** (a textbook gene signature, a published clock) yields a zero-parameter prior that nearly matches a fully trained model and gives the programme's only clean super-additive gain. Finally, gating genome-wide data on a textbook prior and mining the *residual* turns the integrator into a **discovery** tool: it recovers a verified basal/lineage axis beyond "LumA/B = proliferation" (keratinization, $p \approx 8 \times 10^{-11}$) and a verified neuroendocrine/immune axis beyond the ERBB2 amplicon, while correctly finding nothing where the textbook is already complete (ER). Inverting the frame, a *hypothesis* can itself be expressed as a candidate anchor and tested against the textbook prior; a commonality/mediation decomposition then separates a genuinely novel axis from one that is merely redundant (collinear) or absent, and a five-endpoint × four-cohort panel (TCGA RNA-seq, TCGA Agilent, METABRIC, SCAN-B) shows anchor-orthogonal axes transport across platform and cohort while collinear ones do not — their verdict is governed by a nuisance correlation that can flip sign with the assay (RNA-seq vs microarray). The method is modality-agnostic and, above all, honest: discovery where warranted, silence where not, and a quantified account of what fails to reproduce and why.

1. Background

The prevailing assumption that adding omics layers improves prediction is contradicted by careful benchmarks: a large TCGA survival study found that adding data types most often *hurt* performance and that mRNA (±miRNA) sufficed for most cancers (Li et al., 2024); late-fusion work shows the dominant modality is task-specific and pan-cancer multi-omics barely exceeds the best single view (Nikolaou et al., 2023); and naive concatenation routinely underperforms structured fusion (Makrodimitris et al., 2023). The honest goal is therefore not "always fuse" but **never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot.** This note defines an integrator built to that contract and then repurposes it as a discovery engine.

2. Method

All functions live in `omniomics.multiomics`.

Anchor selection (`select_anchor`). Score each modality by repeated stratified-CV performance (primary), tie-break by robustness (subtract a multiple of the CV s.d., weight by coverage). The anchor is the top composite, chosen empirically per task inside the outer CV.

Gated residual (`anchored_gate` / `anchored_integrate`). Pin the anchor; add each secondary on the anchor's residual with a non-negative weight β chosen on held-out data, β = 0 allowed. The combined score is `logit(anchor) + β·secondary`. β = 0 ⇒ combined ≡ anchor (never below); β > 0 only where a secondary earns a margin gain. A constrained, interpretable special case of cooperative learning (Ding et al., 2022).

Forward gating (`forward_integrate` / `auto_integrate`). For ≥ 3 modalities, add one at a time onto the current residual in rank order, keeping each only if it clears the margin — forward selection over modalities, with an interpretable record of which entered and at what β.

Knowledge anchor (`knowledge_anchored_integrate` / `signature_score`). The anchor need not be a data modality: a fixed textbook signature or published clock (zero trained parameters) becomes the anchor and the data is gated onto its residual. The gate then answers a clinically meaningful question — *does the genome-wide data add anything beyond established biology?*

Residual discovery (`anchored_residual_discovery`). Anchor on the prior, gate the data, test that the residual is signal-not-noise, and surface the features driving it that are *orthogonal* to the anchor — candidate biology beyond the textbook — with a matched-random-panel null as the noise control.

Small-sample robustness throughout: a secondary is added only if a non-zero β beats the anchor by more than a `gate_margin`, with each β's inner-CV score averaged over repeated splits.

3. Results (TCGA-BRCA)

Anchored integration never falls below the leader. On LumA/B, RNA alone scores 0.947 and methylation 0.745; a naive stack dips to 0.941 (below RNA) while the gated combiner stays at 0.947 (β = 0 in 9/10 repeats). A constructed positive control — a held-out methylation set RNA cannot see — flips this: the gate engages (β = 4–8) for a significant +0.047. So the gate is both protective and capable.

Not RNA-biased. Run blind on nine expert-defined subtype labels, `auto_integrate` anchors on methylation for all five methylation-defined clusters and on RNA for all four PAM50 calls — anchor selection follows the biology, never below the leader on any of the nine.

Knowledge anchoring — the first clean super-additive gain. A 20-gene proliferation index with zero trained parameters scores 0.919 on LumA/B (nearly the trained 1500-gene RNA model, 0.942); gating genome-wide data reaches **0.947 (Δ +0.029)**, beating the pure data model. With the Horvath clock as the anchor for normal-tissue age the clock alone (0.947) already beats RNA (0.911) and data adds ≈ 0 — the textbook suffices and the gate says so.

Residual discovery, verified. Mining the proliferation-anchored residual on LumA/B surfaces the basal/ squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial r ≈ 0.46). It is verified three ways: a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses (real +0.021 > all permutations). g:Profiler confirms a coherent program — cornified-envelope formation / keratinization / epidermis development (Reactome $p \approx 8\times10^{-11}$).

Generalises, reproduces, and discriminates.

endpoint	textbook anchor (0 params)	anchor AUROC	Δ data adds	verdict
LumA vs LumB	proliferation index (20 genes)	0.919	+0.029	incomplete → basal/lineage axis (verified)
HER2 status	ERBB2 amplicon (9 genes)	0.752	+0.054	incomplete → neuroendocrine/immune axis (verified, ≠ basal)
ER status	ER/luminal signature (20 genes)	0.938	−0.001	complete → no hidden axis (specificity)

The HER2 discovery is a second, *different* verified axis (held-out +0.068 in 8/8 splits; panel p = 0.024; label-specific), a diffuse neuroendocrine/secretory + immune program distinct from basal. ER is a real-data specificity negative: the textbook signature already matches genome-wide RNA.

Modality-agnostic. The discovery runs unchanged on DNA-methylation features; on LumA/B it correctly finds no methylation axis (random and basal-targeted CpGs both Δ ≈ 0) — the basal axis is transcriptional, not predictively methylation-encoded here.

External validation (independent cohort). The basal discovery reproduces in METABRIC (microarray, n = 1175 LumA/B): the fixed TCGA-discovered panel adds Δ +0.036 over the proliferation prior (combined 0.960; beats random panels, p = 0.048), and an unbiased re-discovery in METABRIC independently recovers the same basal genes (KRT5/14/17/6B, TP63, SOX10, DSG3/DSC3 …), overlapping the TCGA panel 20/30 (hypergeometric $p \approx 7\times10^{-27}$). The discovered axis is a reproducible biological signal, not a single-cohort artefact. The HER2 axis, by contrast, does *not* replicate in METABRIC — there the amplicon anchor is near-complete (0.997 vs 0.752 in TCGA), leaving no residual (panel Δ ≈ 0, re-discovery overlap 2/30). Reproducibility tracks biological coherence: the pathway-enriched basal axis reproduces; the diffuse HER2 axis was a cohort-specific residual.

Hypothesis-as-anchor. The frame also inverts: a *hypothesis* is expressed as a candidate anchor and gated onto the textbook anchor's residual for a three-way verdict — SUPPORTED (adds beyond the textbook), EXPLAINED_BY_TEXTBOOK (predicts alone but redundant once the dominant prior is controlled), or REFUTED — operationalising the observation that most signatures predict outcome and a hypothesis is novel only if it survives adjustment for the dominant prior. On LumA/B against the proliferation anchor, a basal/keratinization hypothesis is SUPPORTED (Δ +0.039), immune is REFUTED, and a random gene set is EXPLAINED_BY_TEXTBOOK (predicts at 0.63 alone but adds ≈ 0). Scaled across the 50 MSigDB Hallmark sets, the screen is self-validating — proliferation hallmarks (E2F/G2M/MYC) add exactly 0 beyond the anchor — and, requiring agreement across two proliferation anchors plus FDR, the robust SUPPORTED hits are the estrogen-response lineage programs (matching LumA/B biology). This pathway-level ranking is cohort/platform-sensitive (it does not reproduce in METABRIC microarray, where the anchor-family agreement filter correctly nulls proliferation hallmarks instead), so it is a within-cohort hypothesis-ranking tool, not a cross-platform claim — unlike the gene-level basal discovery, which reproduces. To keep this honest, `hypothesis_anchor_test` returns a gate-free commonality/mediation decomposition (`collinearity_label` ∈ {NOVEL, REDUNDANT, INERT}) that separates an *absent* hypothesis from a merely *redundant* (collinear) one: the estrogen-response axis is NOVEL in TCGA (unique R^2 0.038) but REDUNDANT in METABRIC (redundancy 1.00, 96 % mediated through proliferation). A controlled transportability sweep — fixing both marginal effects and varying only the anchor–hypothesis correlation — shows the *same* ER effect is 100 % NOVEL at TCGA's correlation (+0.19) yet collapses into the collinear/suppression valley at METABRIC's (−0.17): the verdict is a covariate-distribution (transportability) property, not a change in biology. A five-endpoint panel across four columns (TCGA RNA-seq, TCGA Agilent microarray — the same patients on a different platform — the independent METABRIC microarray cohort, and SCAN-B, a fully independent RNA-seq cohort of ∼3,400 tumours) makes the payload concrete: three endpoints transport and two do not. The transportable ones are orthogonal axes (basal→immune and HER2-subtype→immune are NOVEL in all four; ER-status→proliferation carries a small unique slice in all four — robust across platform and independent cohort), while the two non-transporting ones are exactly the ER-collinearity cases (proliferation→ER on Luminal A/B; amplicon→ER on HER2 status). For Luminal A/B the label tracks measurement technology — NOVEL on both RNA-seq cohorts (TCGA RNA-seq, SCAN-B) but REDUNDANT on both microarrays (TCGA Agilent, METABRIC), flipping even on the same TCGA patients. So the caveat attaches specifically to hypotheses whose anchor-collinearity is itself measurement-dependent — anchor-orthogonal axes transport cleanly across platform and cohort.

4. Related work

The approach instantiates three established traditions: clinical-offset / incremental-value modelling, where an established model is an offset and omics are selected on the residual (Volkmann et al., 2019, a near-exact methodological twin); biologically-informed models that bake known biology into structure (Elmarakeby et al., 2021, P-NET; Liu et al., 2024, Pathformer); and ML on established signatures (Sarafidis et al., 2024). Our contribution is the packaged, **never-below-anchor, margin-gated** residual on a zero-parameter prior, plus the residual-discovery framing with a matched-random-panel noise control.

5. Limitations

Predictive gains over a strong anchor are modest; the method's value is *routing and discovery*, not large AUROC wins. Genuine super-additive multi-omics gains are rare on real endpoints (one constructed positive control; one real knowledge-anchored gain). The basal discovery is externally validated in METABRIC (replication + independent re-discovery); the HER2 axis tested negative there (cohort-specific — the amplicon anchor is near-complete in METABRIC), so external reproducibility tracks biological coherence. Discovered axes are candidate hypotheses, not causal claims.

6. Conclusion

Anchor on the leading view (or the textbook), add the rest as a gated residual, and let the gate tell you — honestly — whether the genome-wide data beats established biology. Where it does, the same residual names the new axis (basal for LumA/B; neuroendocrine/immune for HER2); where it does not (ER), the method stays silent. Discovery where warranted, silence where not.

References

Li et al., *BMC Med Inform Decis Mak* 2024 · Nikolaou et al., *Cancer Res* 2023 · Makrodimitris et al., *Brief Bioinform* 2023 · Ding, Li, Narasimhan & Tibshirani, *PNAS* 2022 · Volkmann et al., *BMC Med Res Methodol* 2019 · Elmarakeby et al., *Nature* 2021 · Liu et al., *Bioinformatics* 2024 · Sarafidis et al., *ESMO Open* 2024 · Morandini et al., *GeroScience* 2023 · Horvath, *Genome Biology* 2013.

Code, runners, recorded metrics and CI guards: the `omniomics-prototype` repository. Full method note: `reports/anchored_integration_methods.md`.